

Hands-on Assignment 5

Due Date: See Web

Objective: Design and train a vanilla RNN to recognize palindrome numbers with up to 7 digits.

Palindrome numbers are numbers that read the same forwards and backwards. Here are few examples:

121, 1331, 12321, 44444, 98789, ...

Design your model starting from the code for Tutorial 6. Add an embedding layer. Generate the training data by yourself. Train 3 models with 200, 1000 and 50000 training examples respectively, each with 50% positive and negative examples.

The TA will test-run your models on a test set named “test.csv” in the csv format:

```
123,0
121,1
13543,0
13531,1
```

For a self-challenge, you can use LSTM instead of vanilla RNN. However, no extra points will be given so as NOT to put pressure on others.

Include a README file in your submission, which should contain:

- Training log: hyperparameter and training loss figure,
- Your own testing results, and
- Instructions to run your code on “test.csv” (for all three of your trained models).

Grading Scheme:

- Efforts (1.5 points): Did the student complete all assigned tasks?
- Results (1 point): Performance of model
- 0.5 points: Clarity of instructions and organization of the readme file

Notes:

1. **Submission format:** A single zip file with all your files. No need to include the dataset. The zip file should be named Student_ID_Assign#.zip, i.e. 1234567_Assign5.zip.
2. **headers:** Add the following headers at the beginning of your submission code:

"""

Student Name:

Student ID:

Student Email:

3. If you need help or made mistakes on submission, please email TAs directly.
4. Similarity penalty will not be applied to this assignment. However, the Turnitin report will be used as clues for manual plagiarism detection.

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